

Multi-Source Candidate Data Transformer

Technical Design — Step 1

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1. Pipeline



- **Extract** — parse each source's native format into a common internal shape (resume PDF text + link annotations, AI-structured JSON, header-normalized CSV row).
- **Normalize** — **names, emails, phones (E.164), dates (YYYY-MM), country (ISO-3166), and skills** into one vocabulary — before any merging.
- **Resolve Identity** — match the same candidate across sources by email → phone → normalized name.
- **Merge** — combine matched records using source reliability + per-field confidence (§3).
- **Project & Validate** — a runtime config reshapes the canonical profile, then the output is schema-validated (§4).

2. Canonical Schema

Identity & Contact	Background	Profile & Meta
• candidate_id • full_name • emails[] • phones[] — E.164	• location {city, region, country — ISO-3166} • headline • years_experience • links {linkedin, github, portfolio, other[]}	• skills[{name, confidence, sources[]}] • experience[{company, title, start, end, summary}] • education[{institution, degree, field, end_year}] • provenance[{field, source, method}] • overall_confidence

Normalization: emails lowercased + deduped · phones → **E.164** (unparseable → null) · dates → YYYY-MM ("Present" → null end) · country → **ISO-3166 alpha-2** via direct + state-level lookup, never guessed · skills → canonical alias map ("JS" → "javascript", "Node" → "node.js").

3. Merge & Conflict Resolution

- **Match key** — email (primary) → phone → normalized full name.
- **Source reliability** — Recruiter CSV > ATS JSON > Resume > GitHub > LinkedIn > Notes — used as a tie-breaker weight, not an absolute override.
- **Conflicts** — **highest-confidence source** wins; confidence is lowered and the losing source stays visible in provenance, never silently dropped.
- **Agreement** — two sources producing the same normalized value boosts confidence above either source's baseline.
- **Arrays (emails/phones)** — unioned + deduplicated, not reduced to a single winner.
- **Overall confidence** — **weighted average** of all populated field confidences — thin or conflicting data scores honestly lower.

4. Runtime Custom-Output Config

The canonical profile stays immutable. A **Projection Layer** reads a runtime config and, per field: selects the source path (optionally renamed via **from**), applies a declared normalization, and toggles provenance/confidence. Each field is then type-validated before the response returns. Missing values follow **on_missing**: null (default), omit, or error. Since projection only reads the canonical record, one merge run can produce many differently-shaped outputs with zero re-extraction.

5. Edge Cases & Descopes

- 1.Missing source — remaining sources still process; unavailable fields resolve to null, run still completes.
- 2.Invalid/malformed source — skipped + logged, not crashed; profile built from whatever did parse.
- 3.Conflicting phone/company values — highest-confidence value kept, loss recorded in provenance.
- 4.Duplicate skills, different spelling — canonicalized and merged into one entry with a combined source list.
- 5.Partial data (only CSV or only resume) — valid profile still produced; absent fields stay null, never guessed.

Descoped under time pressure: live GitHub/LinkedIn APIs and ATS-JSON ingestion (Resume + CSV already satisfy "one structured + one unstructured"); fuzzy/NLP entity matching beyond email → phone → name; OCR for scanned resumes; sub-month-precision years_experience.